

Signal Optimization as a Safety and Operation Countermeasure

Advisor: Dr. Fahmida Rahman



Project Goal

- Evaluate whether existing signal timing at signalized intersection(s) contains inefficiencies or latent safety risks and conduct a literature review to identify optimization strategies using field-collected datasets.

Key Tasks

- Collect existing-condition field data in NJ
- Identify ineffective or “wasted” green time
- Diagnose hidden operational and safety risks
- Do a Literature Review for signal optimization strategies
- Evaluate signal optimization as a safety countermeasure emphasizing low-cost, no-infrastructure solutions

Clinic Impacts:

- Industry Software Skills: VISSIM
- Professional Connections: ASCE/ITE Conference, Industry Networking